

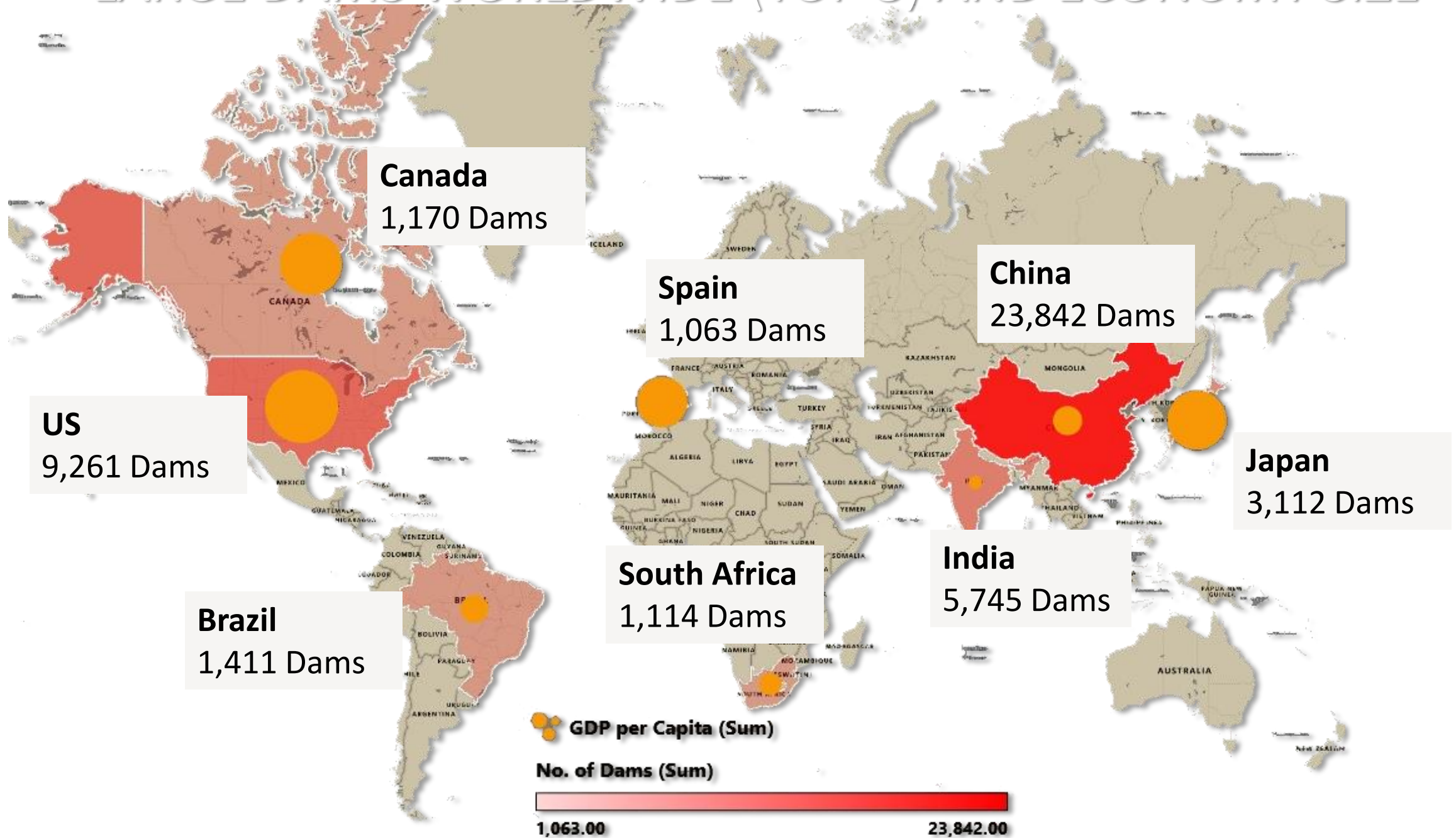
RESERVOIR OPERATION AND FLOOD MANGEMENT

Status and Challenges
in India

Pramod Narayan,
Project Director, DRIP



LARGE DAMS WORLDWIDE (TOP 8) AND ECONOMY SIZE



INTEGRATED FLOOD MANAGEMENT APPROACH

- Complete immunity not possible – can be partially controlled
- Impacts can be minimised – managing to reduce losses

IFM



STRUCTURAL MEASURES

- Physical in nature
- Aim at preventing flood waters from reaching potential damage areas

NON-STRUCTURAL MEASURES

- Strive to keep people away from flood waters
- These use knowledge, practice and agreements to reduce the risk and impacts through training, awareness and capacity building.

Modify Flood
Magnitude

Living with floods- Planned
Activity to Modify susceptibility
to Flood Damages

STATUS OF RESERVOIR OPERATION DURING FLOODS

- Few large dams had **well defined** O&M as well as EAP protocols.
- **Flood management** aspects given **least weightage** in reservoir operations.
- Current state of practice has raised fears of **dam induced floods**.
- Inadequate **O&M budget**
- Lack of **skilled dam staffs** and long term capacity building.

The onset of the monsoon has raised fears of dam-induced floods as India's major reservoirs are already full. Live storage in 123 reservoirs was 55.837 billion cubic metres (BCM), according to data released by the Central Water Commission (CWC) on June 4, 2020.

Live storage in reservoirs in the corresponding period last year was 32.801 BCM, with the average for the last 10 years at 33.706 BCM.

The current live storage available in 123 reservoirs was 170 per cent of the same corresponding period of last year and 166 per cent of the last 10 years' average.

This does not bode well for flood management in a country where extreme rainfall events are becoming a norm.

It is, thus, crucial to carefully manage the release of water from dams, as the monsoon reaches the rest of the country in the coming days.

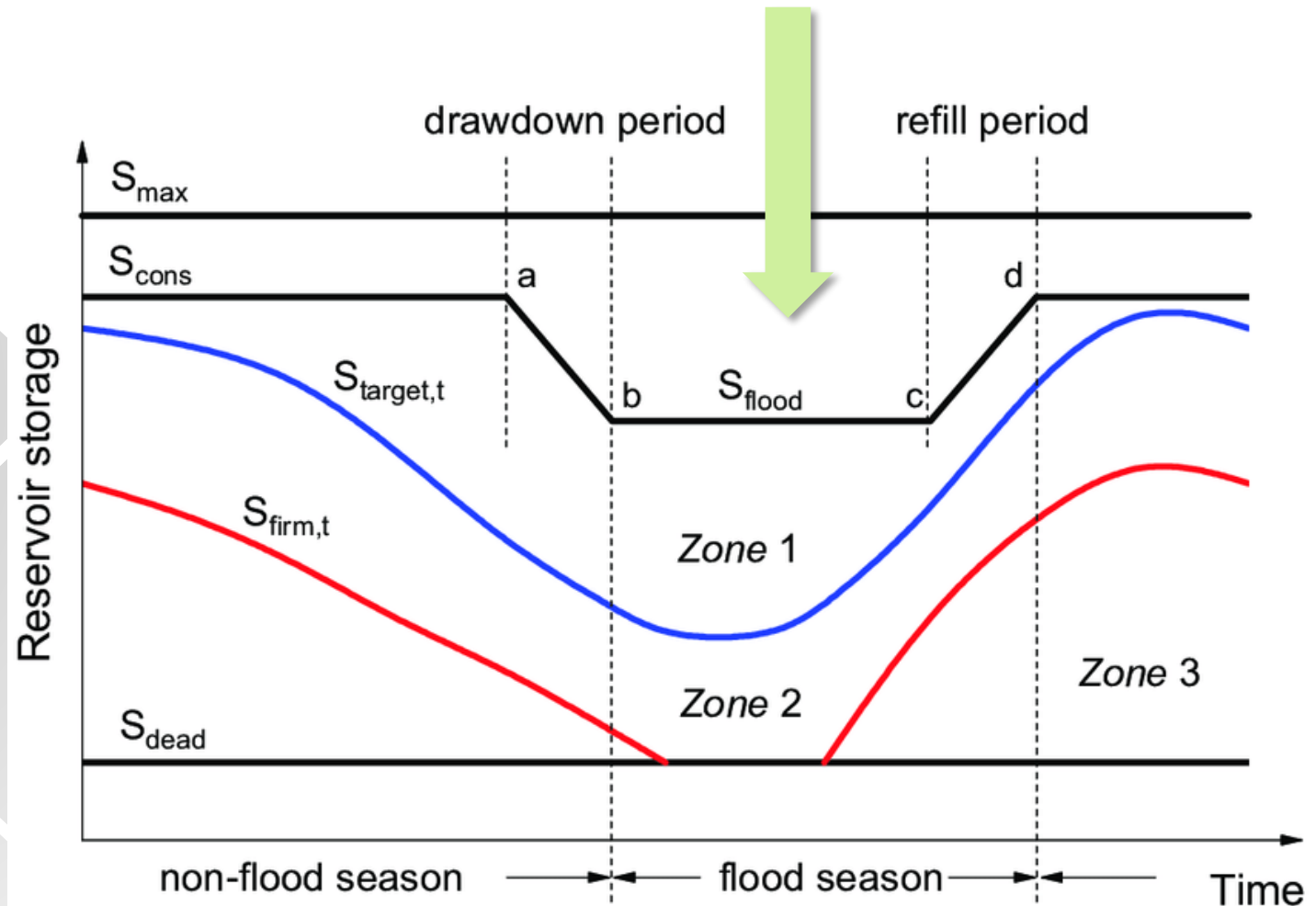
The rule curve — an international practice of dam operation — must be followed by dam management authorities, as any lapse may lead to dam-induced floods, of which there are several instances in the

FLOOD MODERATION

- Only few Reservoir(s) has dedicated flood cushions to moderate flood

- For reservoir without flood cushion → Rule curves (specifying levels to be maintained in the reservoir at different times in a year) of major **reservoirs where flood cushion is not in-built need to be reviewed** to have some dynamic flood cushion for major part of the flood season.

Dynamic flood Cushion



CHALLENGES



AGEING IMPACTS

Adds vulnerability to

- Structural stability
- Seepage
- Material deterioration
- Internal Erosion and piping

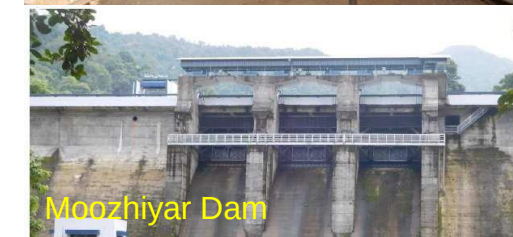
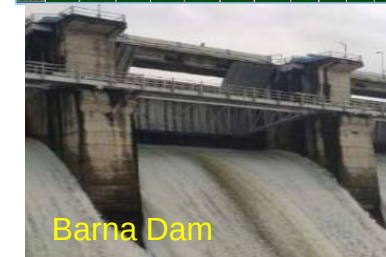
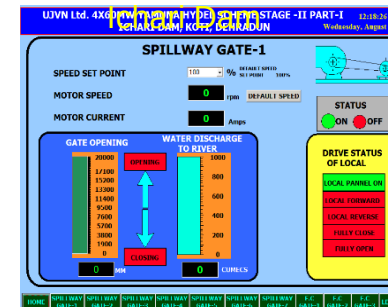


Timely O&M and
Refurbishment of Dam in
general Address this
problem.



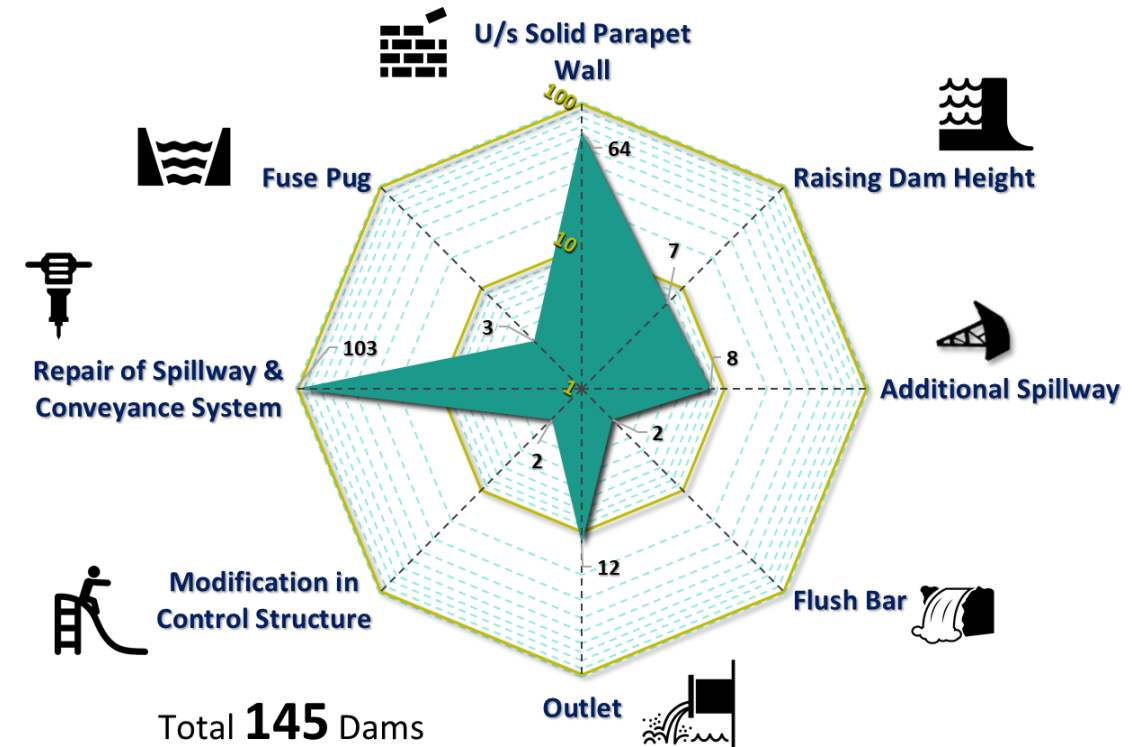
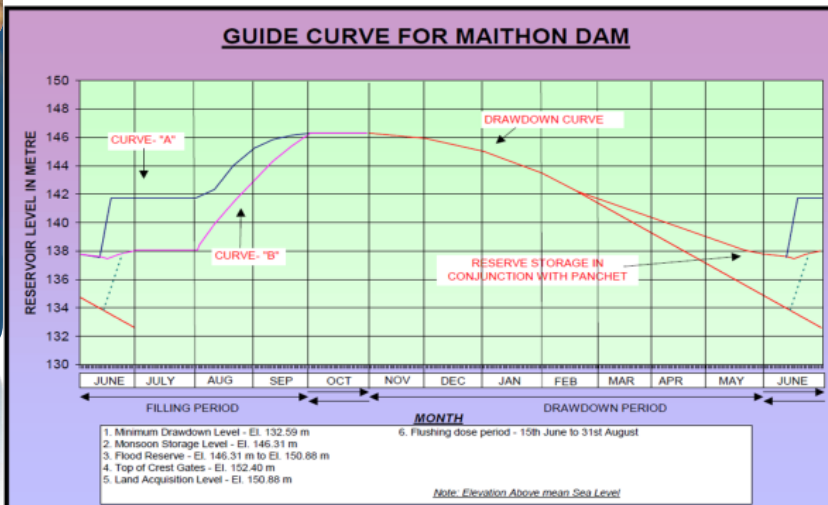
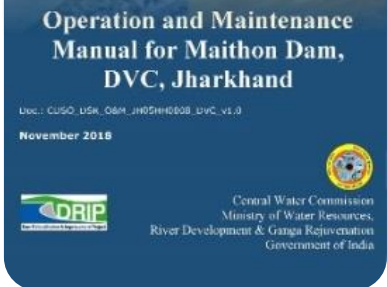
OPERATIONAL SAFETY

- ❑ One of three Safety of a Dam, deals with hydro-mechanical (HM) equipments and associated components
- ❑ During Extreme Flood, Reservoir Shall be Operated Safely and Efficiently
- ❑ Need Regular Review Keeping in view Continuous Wear & Tear of these Components, Increase in Design Flood may result in increase in Water level coupled with increase in hydraulic loads
- ❑ Under DRIP, rehabilitation also include spillway gates, hoisting arrangements, Gantry Crane, Control equipments etc., improved operational performance of 147 dams located in seven States.



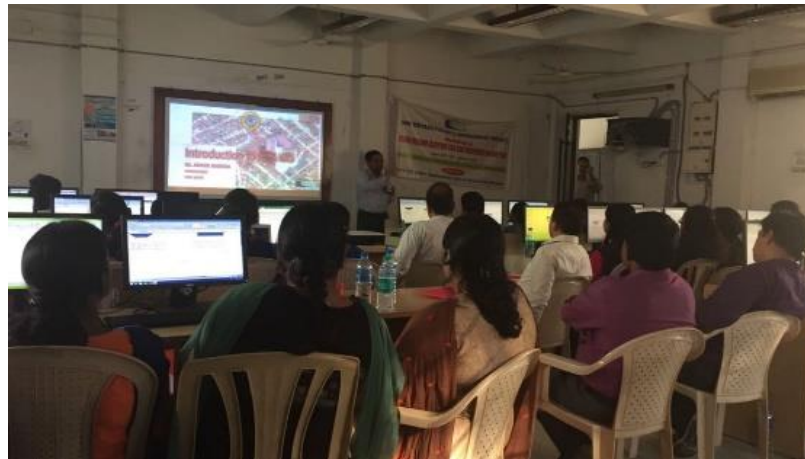
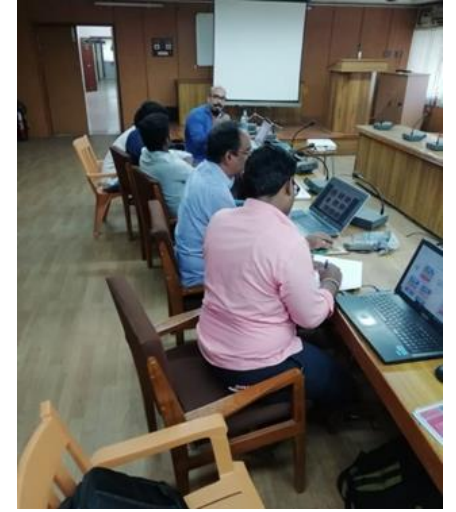
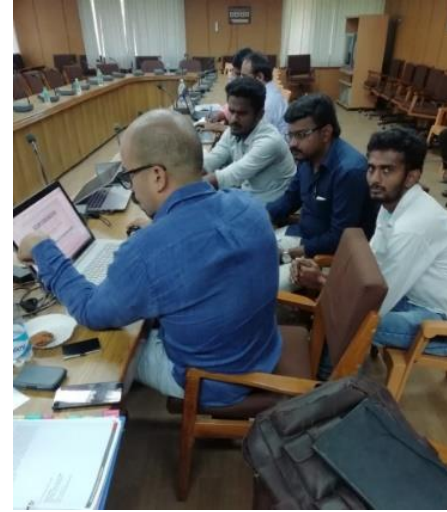
HYDROLOGICAL SAFETY

- Hydrology is a Dynamic Science, hence the Design Flood
- Design Flood Review of 207 reservoirs
- 32 Reservoirs($IDF \leq ODF$), 175 Reservoirs ($IDF > ODF$)
- 115 Reservoirs, IDF increased by more than 50%; 68 reservoirs IDF increased by more than 100% of the original design flood.
- Hydrological safety addressed through **structural** and **non-structural** measures



INSTITUTIONAL STRENGTHENING

- ❑ Fuel of Vehicle to Reach Destination
- ❑ Continuous Process
- ❑ Continuous Improvement in Skill and Knowledge of Staff
- ❑ Collaborations with Leading Global Institutions for knowledge & Technological Advancement
- ❑ Customized Training Programs
- ❑ Public Outreach Programs
- ❑ Partnering with Apex Indian Academic and Research Institutions
- ❑ Workshops/Seminars/Conference
- ❑ Tailor Made Schemes



CLIMATE CHANGE CHALLENGES

New dams and reservoir required in order to mitigate climate change by equilibrating the perturbed water resources regimes and to ensure water, energy and food as well as to protect against floods.

Retrofitting of Existing Dams at a marginal cost to create Flood Cushion, Hydel Power, Additional Storage to mitigate Drought



DAM SAFETY LEGISLATION

To provide for surveillance, inspection, operation and maintenance of the specified dam for **prevention of dam failure related disaster** and to provide for **institutional mechanism** to ensure their safe functioning and for matters connected therewith or incidental thereto



Complete, comprehensive, well-structured legislation aligned with the world's best practice

Passed by Lok Sabha on August 2, 2019

To be introduced in Rajya Sabha for discussion

Roles are assigned, in particular, separation between Owner/Operator and Regulator

Institutional structure envisioned in the Bill

I: Preliminary

II: National Committee on Dam Safety

III: National Dam Safety Authority

IV: State Committee on Dam Safety

V: State Dam Safety Organisation

VI: Duties and functions in relation to dam safety

VII: Safety, inspection and data collection

VIII: Emergency action plan and disaster management

IX: Comprehensive dam safety evaluation

X: Offences and penalties

XI: Miscellaneous

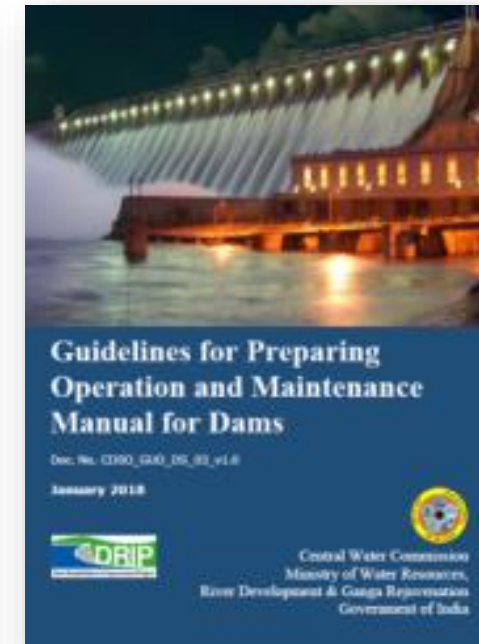
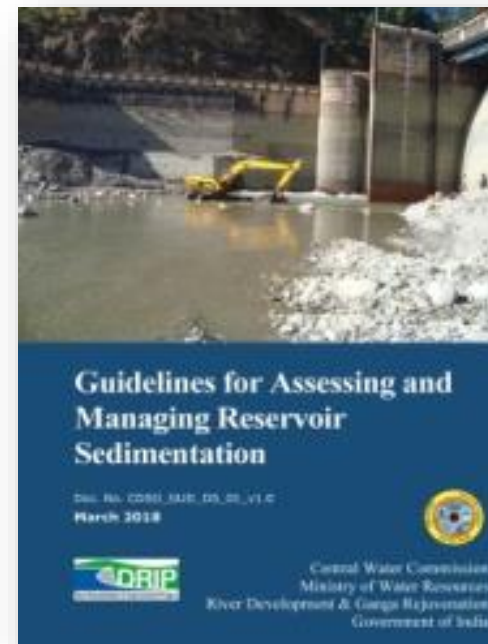
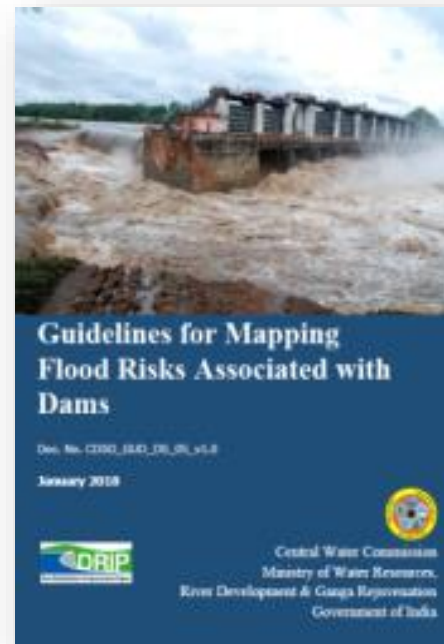
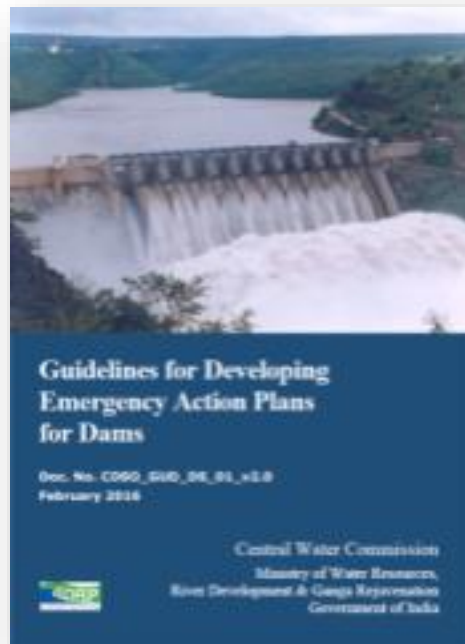
1st schedule: Functions of National Committee on Dam Safety

2nd schedule: Functions of National Dam Safety Authority

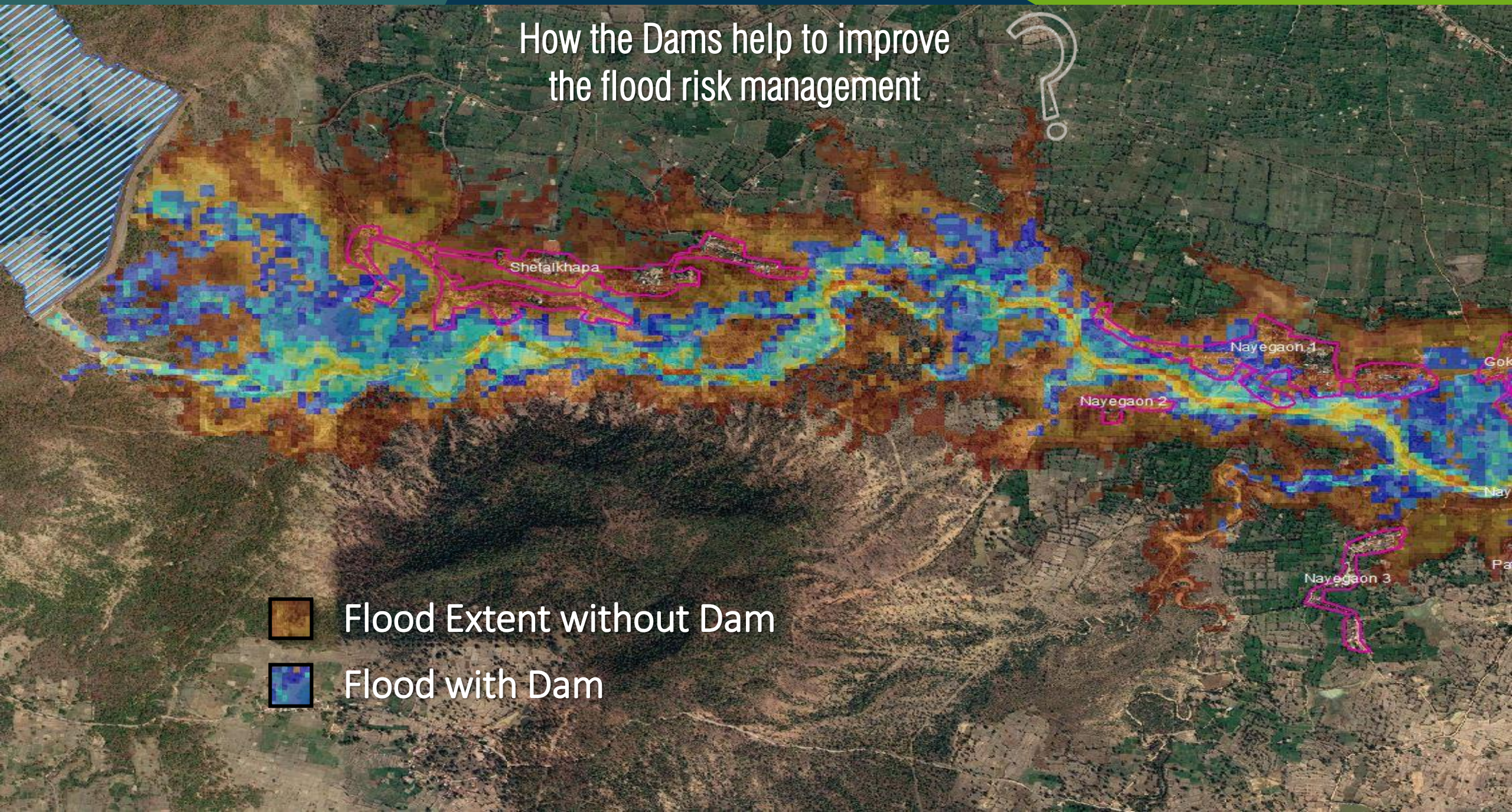
3rd schedule: Functions of State Committee on Dam Safety

TECHNICAL REGULATIONS

- Technical Regulation is one of important Pillars
- Uniformity and Standardisation of Various Practices
- 13 Guidelines and Manuals published under DRIP



How the Dams help to improve the flood risk management



Flood Extent without Dam



Flood with Dam

FLOOD RISK MANAGEMENT



What are the hazards and how likely are they to occur?

How will the infrastructure perform in the face of these hazards?

Who and what are in harms way? How susceptible to harm are they? How much harm is caused?

Risk = f (Hazard, Performance, Consequences)

FLOOD RISK MANAGEMENT



- ① What Can Be Done About the Risks?
- ② What Does this Mean for the Community?
- ③ What is the Source of the Risks?
- ④ Compared to Others, How Significant is the Risks?

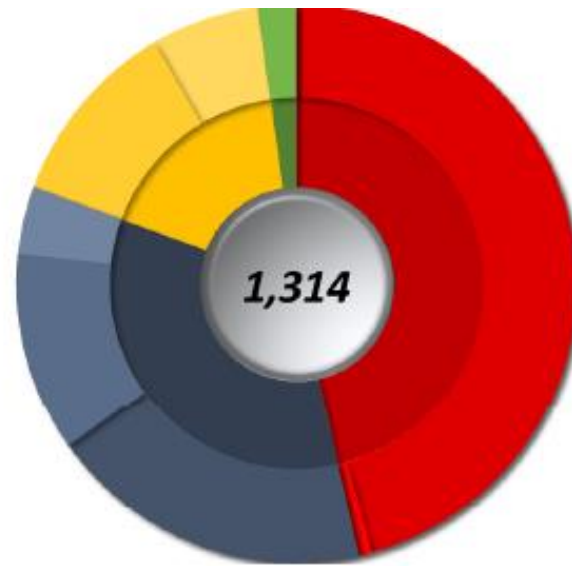
**Risk = f (Hazard, Performance,
Consequences)**

RISK FLOOD MANAGEMENT

$$\text{Risk} = f(\text{Hazard}, \text{Performance}, \text{Consequences})$$

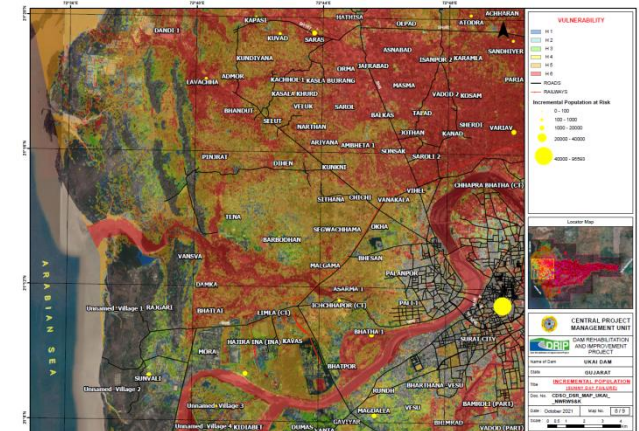
- Risk = f (Hazard, Performance, Consequences)
- Need to Switch for Risk Based Decision System
- Guidelines For Classifying the Hazard Potential of Dams, 4 Lakes of Chennai and other Peripheral Reservoirs Need this

Consequences Category	Indicator Parameter	Consequences Rating Points [1]	Key Remarks
Capital Value of Project	Dam Height (I_{DH})	82	Major Irrigation project with a cultivable command area of 3, 31,577 ha.
	Project Benefits (I_{PB})	143	
Potential for Loss of Life	Population at Risk	600	Major population at risk is concentrated in Surat city in Surat district. Potential for loss of life is assured.
	Critical Arrival Time	10	
Potential for Property Damage	Transportation	52	Major consequences restricted to agricultural land use.
	Infrastructure Damaged	246	
	Services Disrupted	152	
Potential for Environmental & Cultural Impact	Environmental Impact	0	Minimal incremental damage. Short-Term or reversible impact is expected (less than 2 years)
	Cultural Heritage	29	
Total Potential Consequences Index (P_{CI}) =		1,314	
Potential Hazard Class =			Class IV



- Capital Value of Project
- Potential for Loss of Life
- Potential for Property Damage
- Potential for Environmental and Cultural Impact

Ukai dam Hazard Classification
(Class IV – Extreme Hazard)



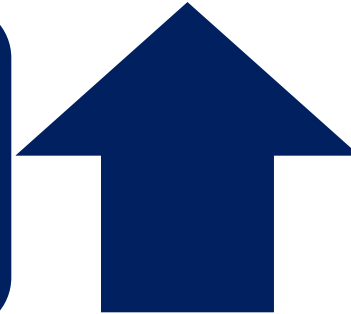
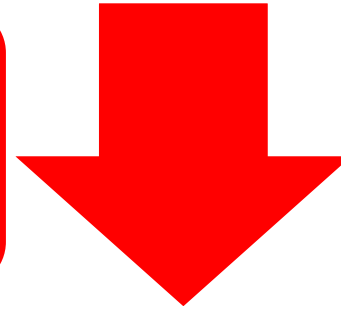
RESERVOIR SEDIMENTATION: A GLOBAL CONCERN

One of the Major Problems for Dam Safety & Water Management

- ✓ ~50% Storage Loss by 2050
- ✓ 100% within 200 to 300 years
(very preliminary in nature)

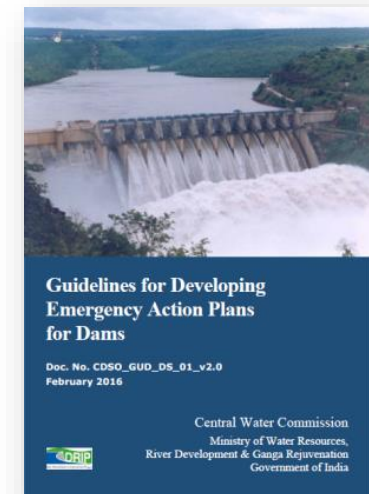
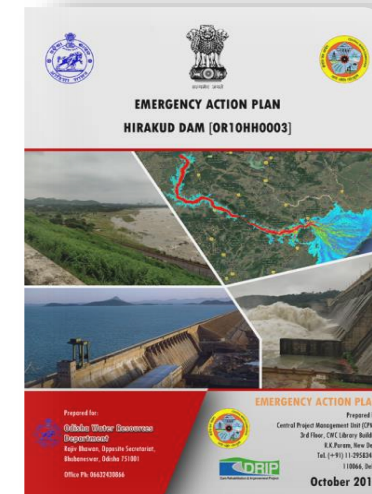
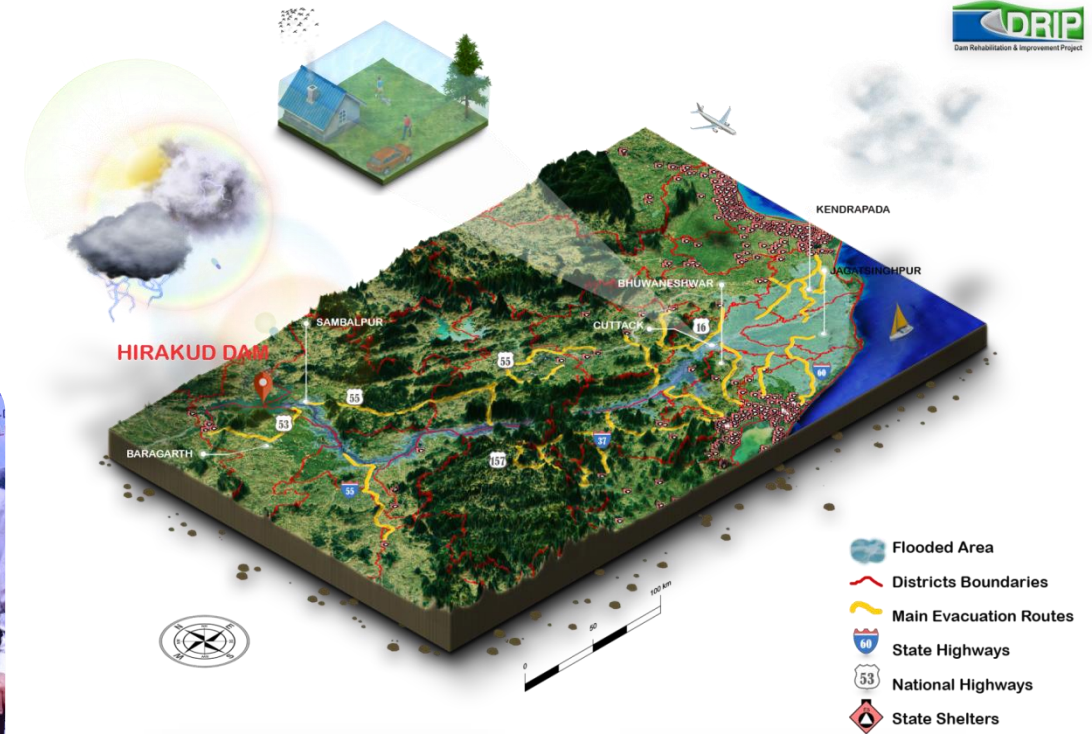
(Reservoir Conservation, Volume I, The RESCON Approach, 2003, IBRD)

- ✓ Water & Energy Demand
- ✓ Flood Management & Risk
- ✓ Safety Concerns

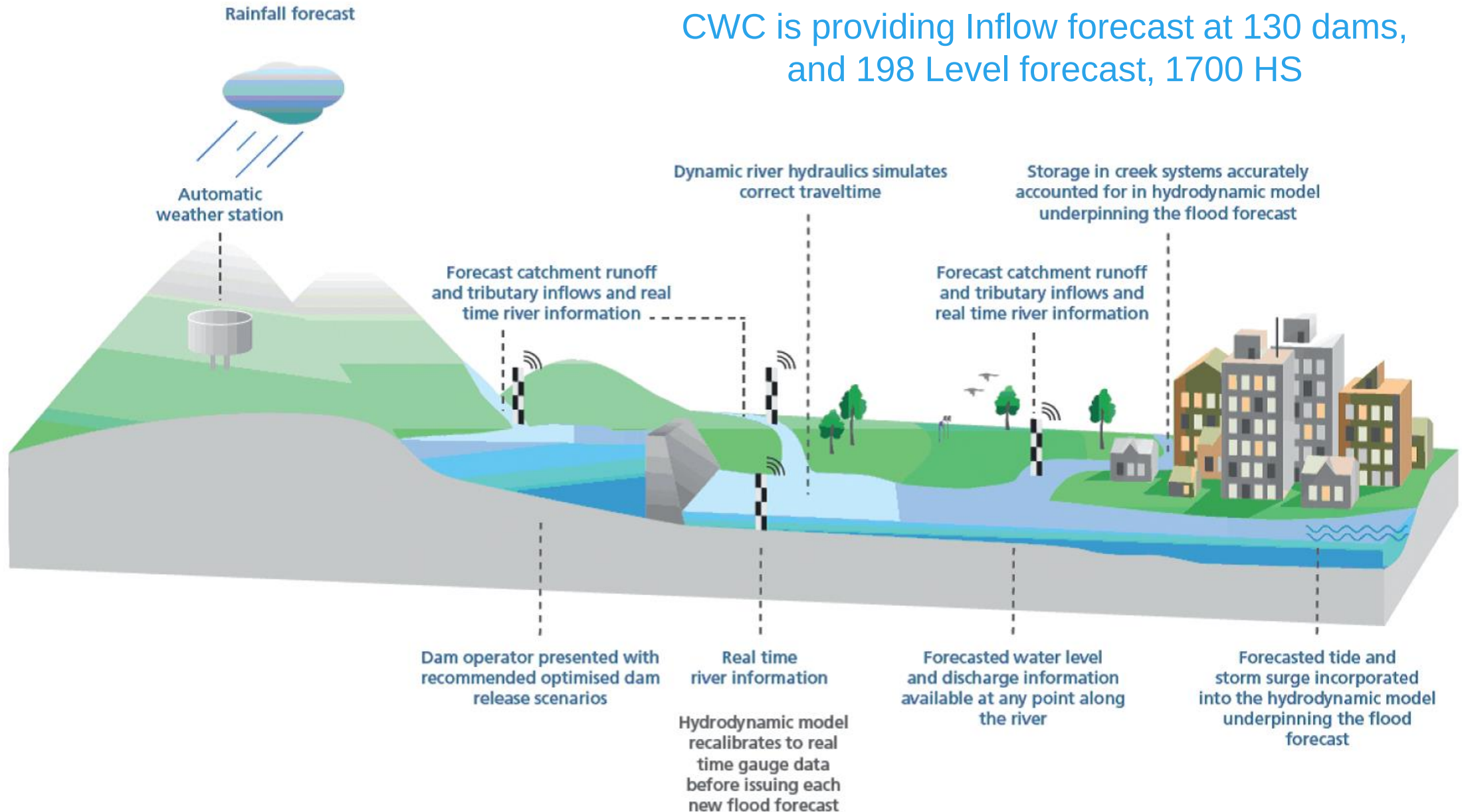


DISASTER MANAGEMENT & PUBLIC AWARENESS

- Dam Break Analysis and Inundation Mappings
- Dam Specific Emergency Action Plans
- 103 Stakeholder Consultation Programs, about 2K Participants



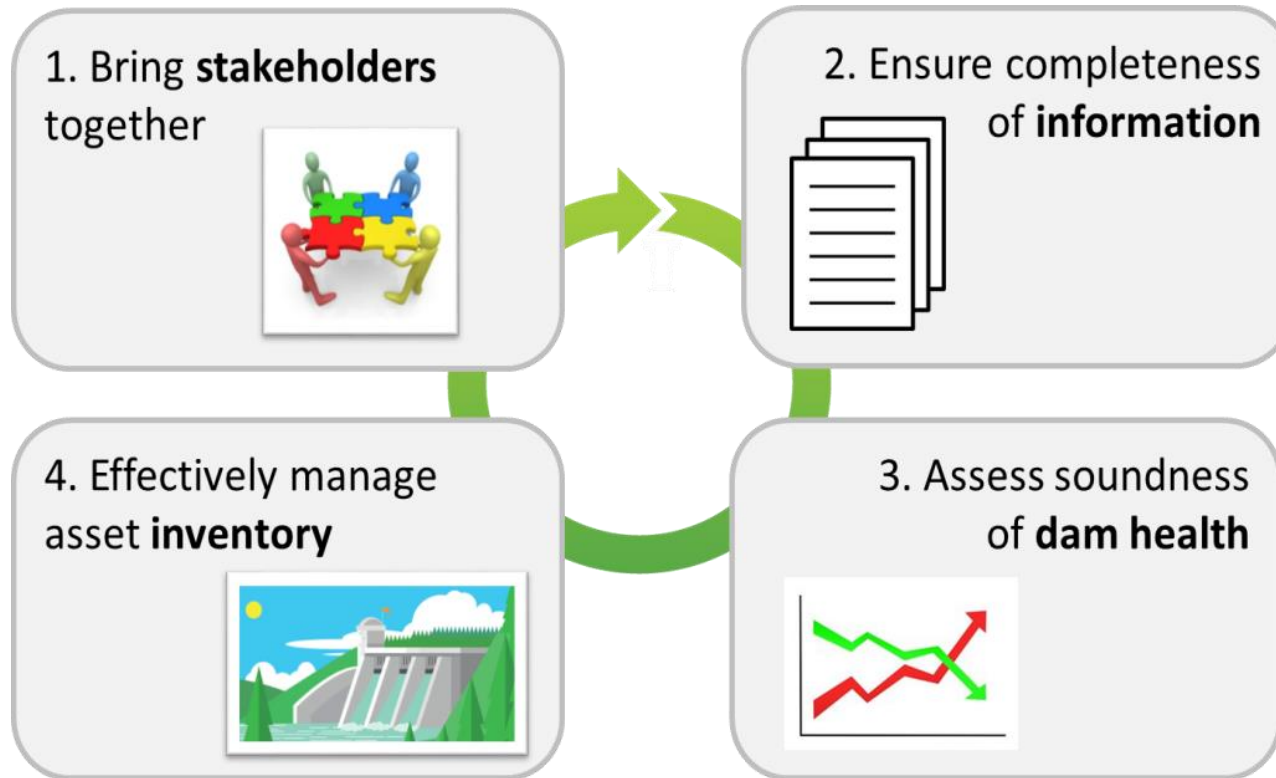
CWC is providing Inflow forecast at 130 dams,
and 198 Level forecast, 1700 HS



ASSET MANAGEMENT



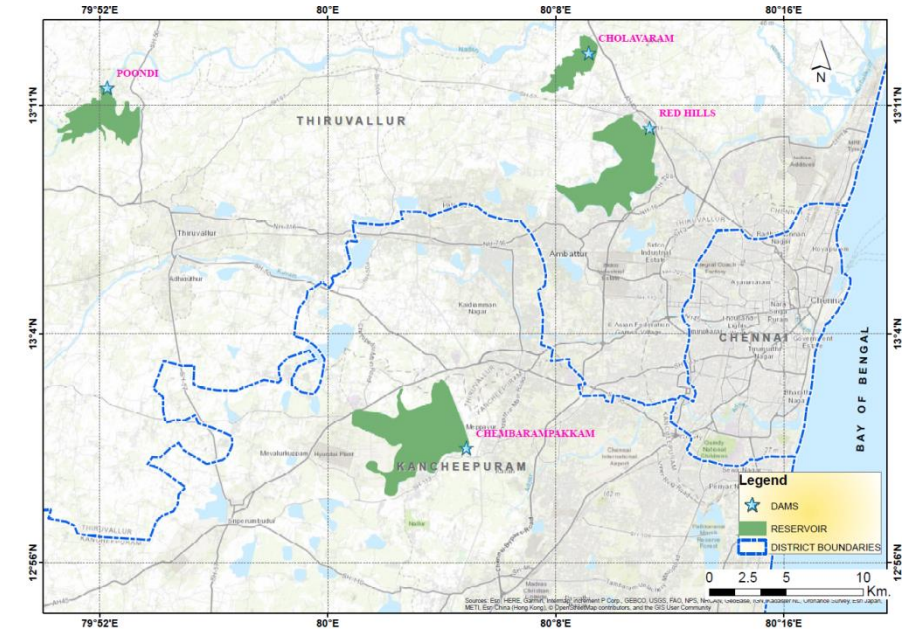
Web-based Asset Management Tool
Repository of Dam Health information for long-term sustainability;



- ✓ Asset Management Essential For Long Term Sustainability
- ✓ Knowledge Transfer From One Generation to Other
- ✓ Storage and Archival of Information Important For Framing Strategic Plan
- ✓ Digital Technology Can be Vital
- ✓ Enhance the Preparedness to minimise Consequences
- ✓ Step to Move For Application of Artificial Intelligence

INTEGRATED FLOOD MANAGEMENT

- ❑ About 75% of Total Water Supply met by 4 Reservoirs of Chennai
- ❑ Flood 2015, Chennai nearly went under water
- ❑ Safeguarding These Assets, Important for reliable water supply and Public Safety
- ❑ DRIP Placed Dam Specific O&M Manual, Emergency Action Plan to Deal With Any Exigency.



DRIP INITIATIVES FOR IFM



Population at Risk

1,633,533 nos. (Chembarapakkam)
2,11,553 nos. (Red Hills)
3,85,709 nos. (Poondi)
1,03,706 nos. (Cholavaram)



Affected Area

270 sq.km (Chembarapakkam)
92 sq.km (Red Hills)
427 sq.km (Poondi)
40 sq.km (Cholavaram)



D/S slope strengthening (Chembarampakkam)



Standardization of Bund (Cholavaram)

Issues For Govt. of TN:

- ☐ Ensure Adequate Annual O&M Budget
- ☐ Explore Generation of Incidental Revenue
- ☐ Water Pricing



EMERGENCY ACTION PLAN CHEMBARAMBAKKAM TANK [GENTNWA001]



Prepared for:
Tamil Nadu Water Resources Department

Prepared by:
Executive Engineer, WRD,
Kosasthalaiyar Basin Division,
Thiruvallur, Tamil Nadu.
Tel. 044-27632779

APRIL 2021



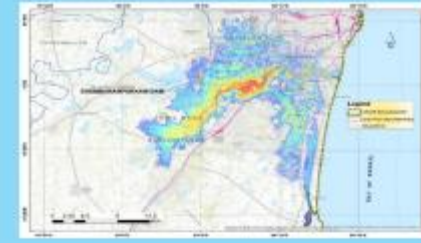
GENTNWA001 OPERATION AND MAINTENANCE MANUAL OF CHEMBARAMBAKKAM TANK



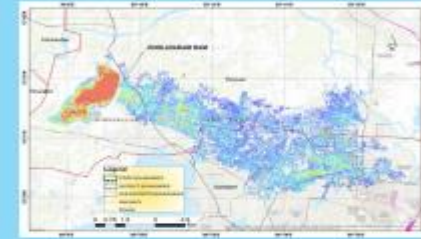
OFFICE OF THE,
EXECUTIVE ENGINEER, PWD/WRD.,
KOSASTHALAIYAR BASIN DIVISION, THIRUVALLUR
March 2021



Central Water Commission,
Ministry of Water Resources,
River Development & Ganga Rejuvenation,
Government of India.



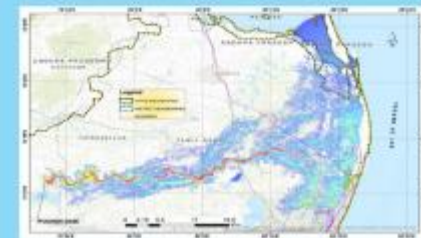
Chembarapakkam



Cholavaram



Red Hills



Poondi

The image features a decorative header at the top with three horizontal bands of color: a dark teal band on the left, a dark blue band in the middle, and a lime green band on the right. On the left side of the slide, there is a large, stylized geometric pattern composed of overlapping rounded squares. These squares are arranged in a triangular formation pointing downwards and are colored in a gradient from dark teal at the top to bright lime green at the bottom. The text "THANK YOU" is positioned on the right side of the slide in a bold, black, sans-serif font.

THANK YOU